Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. Which of the following steps are integral to the process of implementing credit card fraud prevention using IBM Watson? (Select all that apply) 1 / 1 point

- [x] Manually Reviewing Every Transaction
- [x] Monitoring and Fine-tuning the Model

  Nice work

  Correct! Monitoring and fine-tuning the model after deployment ensures its ongoing accuracy and adaptability to new fraud patterns.

- [x] Randomly Sampling Transactions
- [x] Deploying the Model in Production

  Nice work

  Correct! Deploying the model in a production environment enables real-time fraud detection in practical applications.

- [x] Training the Model

  Nice work

  Correct! Training the machine learning model allows it to learn patterns of fraudulent and non-fraudulent transactions, improving its accuracy.

- [x] Data Collection and Preparation

  Nice work

  Correct! Proper data collection and preparation are foundational to building a machine learning model capable of identifying fraud.

2. When assessing the performance of a fraud detection algorithm in a dataset characterized by a significant imbalance between fraudulent and non-fraudulent transactions, which metric is the most effective? 1 / 1 point

- (•) F1 Score
- ( ) Mean Squared Error
- ( ) ROC AUC
- ( ) Accuracy

  Nice work

  Correct! The F1 Score balances precision and recall, making it suitable for evaluating models on imbalanced datasets.

3. Which specific IBM Watson service is designed to support the development and deployment of models for detecting credit card fraud? 1 / 1 point

- ( ) IBM Watson Assistant
- (•) IBM Watson Machine Learning
- ( ) IBM Watson Natural Language Understanding
- ( ) IBM Watson Tone Analyzer

  Nice work

  Correct! IBM Watson Machine Learning provides the necessary tools for building, training, and deploying fraud detection models.

4. What key benefit does the ROC AUC metric provide when evaluating fraud detection algorithms? 1 / 1 point

- ( ) It focuses exclusively on the true positive rate.
- ( ) It is inherently easier to interpret than precision-recall curves.
- ( ) It remains unaffected by class imbalance in the dataset.
- (•) It evaluates both the true positive rate and the false positive rate, offering a comprehensive view of model performance.

  Nice work

  Correct! The ROC AUC metric provides an overall picture by balancing the true positive and false positive rates, which is critical in evaluating fraud detection models where minimizing false positives is important.

5. A financial institution is looking to enhance its fraud detection capabilities using IBM Watson. The data science team needs to evaluate the performance of their newly deployed fraud detection model. Which of the following metrics should they prioritize to ensure a thorough evaluation of the model's performance? (Select all that apply) 1 / 1 point

- [x] Precision

  Nice work

  Correct! Precision measures how many of the predicted fraud cases are actually fraudulent, which is crucial for assessing model accuracy.

- [x] Recall

  Nice work

  Correct! Recall assesses the model's ability to identify all actual fraudulent cases, which is important for detecting as many fraud cases as possible.

- [ ] Log-Loss
- [x] F1 Score

  Nice work

  Correct! The F1 Score combines precision and recall into a single metric, offering a balanced evaluation of the model's performance.

- [ ] Root Mean Squared Error